

Optical Music Generation: Leveraging Off-The-Shelf Diffusion Models to Synthesise Handwritten Scores

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Abstract. The field of Historical Document Recognition faces significant challenges due to the limited availability of transcribed and easily accessible source materials. This issue becomes particularly limiting when tackling the recognition of documents containing elements other than text written in widespread writing systems, where only a few dozen transcribed pages can be assumed to be available. This issue is especially evident in the Optical Music Recognition (OMR) task, even more so when concerning handwritten and historical data. In light of this, in this paper, we explore strategies of Optical Music Generation for compensating the lack of real, annotated data of handwritten music scores by designing synthetic samples. Specifically, we exploit Diffusion Model-based solutions to produce images of scores conditioned on the output of different engraving software. In this way, we can forge a large number of images with known ground truth without requiring specialised data.

Keywords: Optical Music Recognition · Image Generation · Low Resource · Handwriting Recognition

1 Introduction

Music scores are particularly interesting documents to analyse, both for their relevance in the Cultural Heritage domain and for the technical challenges they pose. Recent years have witnessed increasing research attention from the Digital Humanities and the Document AI communities for the development of automatic music recognition strategies. These concern both typewritten music scores — in the Optical Music Recognition (OMR) variant of the task — and handwritten scores — in the Handwritten Music Recognition (HMR) variant.

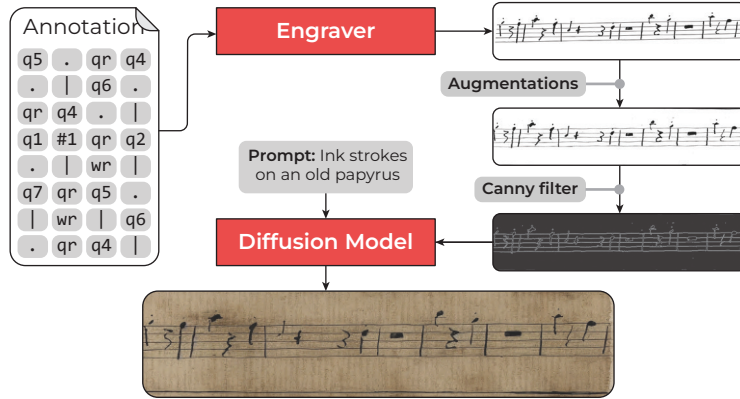


Fig. 1. In this paper we propose a simple pipeline that exploits pre-trained Diffusion Models to produce handwritten-looking images without specific training data. We condition a ControlNet [1] on handwritten-looking images and a prompt that produces an old, handwritten texture.

Perhaps one of the most widely accepted — and discouraging — truths about HMR, however, is the serious lack of annotated data available [23], which limits the range of applicable recognition techniques and their achievable performance. One possible way of circumventing this issue would be generating realistic-looking images of handwritten music samples using image generation techniques, in analogy to the field of Handwritten Text Recognition (HTR) [4]. A recent trend for boosting HTR performance entails training the recognition model on automatically-generated data that mimic the target documents in terms of content and appearance. To obtain these data, Handwritten Text Generation (HTG) strategies have been developed by exploiting large, multi-author HTR datasets such as IAM [5]. However, a similar strategy is not straightforwardly applicable in HMR, given that an equivalent to the IAM dataset in terms of size, availability of transcriptions, and variety of styles does not exist, the closest being the MUSCIMA++ [6]. Thus, the field is stuck in a “chicken-and-egg” situation, where, in order to palliate the lack of data, researchers would need to have that data to begin with.

In light of these considerations, in this paper, we explore a series of strategies to forge realistic-looking, paired data for HMR without requiring any music-specific training data. To this end, we investigate leveraging diffusion models under the assumption that, within their large, generalist training datasets, there are samples of real handwritten documents. We show that using computer-engraved scores as input to a ControlNet model [1] it is possible to produce samples that more realistically look like historical handwritten scores. Moreover, we analyse the limitations and the failure scenarios of this approach to provide guidelines for future works on the topic of Optical Music Generation (OMG) and Handwritten Music Generation (HMG).

2 Related Work

Optical Music Recognition. Research on Optical Music Recognition (OMR) has advanced quite substantially [2], mostly thanks to the adoption of Deep Learning techniques and the appearance of large datasets specialised in music recognition [7, 8, 9, 6]. Many architectures are being developed on typeset scores with competitive results [10] even in complex full-page scenarios [11, 12]. On the other hand, given the lack of large datasets of historical music in Common Western Music Notation, error rates in HMR are still very high, with the current best models sitting over a 25% of Symbol Error Rate [13, 14].

Image Generation. Generating images has long been a field of interest within the Deep Learning community. The technique that has been most widely used to this avail is the Generative Adversarial Network (GAN) [15]. In more recent times, there have been developments in a new paradigm of image generation through Denoising Diffusion Probabilistic Models [16]. Their core idea is to define image generation as a progressive Gaussian Noise removal task. By incrementally adding Gaussian noise on a clean image and training a model to predict the newly added noise signal, one can generate images from noise by iteratively applying the model on a random-valued matrix and gradually producing a structured output. To condition such generation process, Stable Diffusion [17] exploits a text prompt that determines the contents of the generated image. On the other hand, ControlNet [1] allows incorporating additional arbitrary conditioning signals by training separate denoising networks on specific input image domains (*e.g.*, bounding boxes, segmentation regions, contour maps, etc).

Handwritten Text Generation. Handwritten Text Generation (HTG) is used to improve text recognition models with synthetic samples and it involves providing a generative model with some examples of the desired handwriting style alongside a transcription of the text to be rendered. Initially, most research was focused on GANs [4, 18, 19, 20, 21], but more recently, Diffusion Model-based solutions have been proposed [22, 23], which show impressive performance also in terms of the usefulness of the generated samples for enhancing HTR. Unfortunately, most of these techniques are not currently applicable for the generation of handwritten music because there are not enough aligned transcriptions of old scores nor fonts.

Optical Music Generation. Unlike text, the generation of synthetic music scores is quite unexplored. Recently, Lin *et al.* produced a page-level dataset and a first baseline for text-to-score generation [24]. Unfortunately, their pipeline does not allow having fine-grained control over the musical content of the scores. Only Shatri *et al.* [25] have used Deep Learning-based image generation techniques to produce outputs with known, paired transcriptions. The authors employ a GAN-based approach to generate full crops of scores by conditioning on their typeset counterparts. However, this approach is restricted to grayscale images and is not tuned for historical scores. An algorithmic approach to music generation is presented by Mayer *et al.* in [26], where they implement a full engraving system capable of using handwritten glyphs from the MUSCIMA++ dataset [6].

3 Method

In this section, we present our pipeline for generating historical-looking handwritten music score images for which paired transcriptions are readily available. Specifically, given a piece of music content, we rely on an engraving software to obtain a binary image representing it. Then, we obtain a Canny map of this image, which serves as a control signal for guiding the generation with a pre-trained Diffusion Model, alongside a textual prompt to control the style and appearance of the final image. A pictorial representation of our pipeline is given in Fig. 1.

Controlling the Music. The chosen conditioning on Canny obeys the necessity of generating the desired musical content for the scores. Given the absence of transcriptions of handwritten music lines and the fact that most pretrained models are not proficient in music formats, the most approachable technique to explore in the context of HMR is style transfer in an image-to-image setup. We have explored conditioning a ControlNet 1.1⁴ on Canny-filtered versions of computer-engraved scores, for which transcriptions are known in advance or can be generated algorithmically.

Providing a Handwritten-like Source. Given that a ControlNet conditioned on Canny maps follows the contour signal very strictly, the goal is passing images with a style as close to handwritten as possible as input. We employ the first version of the Mashcima [26], which uses cropped musical symbols from the MUSCIMA++ dataset as its font to produce handwritten-looking synthetic music images. We then use the Augraphy data augmentation library [27] to apply ink effects (Ink bleed, bleedthrough) and artifacts (Gaussian noise, page imperfections) and a combination of Augraphy and Albumentations to insert some geometric distortions to the image (ink shifting, affine and elastic transform). The resulting image is then seamlessly blended with a paper-like background for the added texture. After this last step, the Canny map of the image is computed and fed into the model.

Styling. The overall look and feel of the images is controlled via a textual prompt fed into the ControlNet. We explored automatic ways of generating prompts based on real scores via CLIP Interrogator⁵ (reported in [B]), but settled on some handcrafted ones because we found simpler, to-the-point prompts much more stable and easy to get reasonable results from. These prompts are: (0) *An image of an old document*, (1) *An image of an old music document*, (2) *Ink strokes on an empty page*, (3) *Ink strokes on an old papyrus*, (4) *Ink strokes on old parchment*.

Generating odd aspect ratios. Diffusion models can generate higher quality and higher resolution images than their forerunners – Variational Autoencoders and GANs. Nevertheless, the outputs of many of these models’ variants are bound by their training resolution. Generating images at higher resolutions yield sub-optimal results, with flagrant artifacts or repetitive patterns appearing in

⁴ <https://github.com/llyasviel/ControlNet-v1-1-nightly>

⁵ <https://huggingface.co/spaces/pharmapsychotic/CLIP-Interrogator>

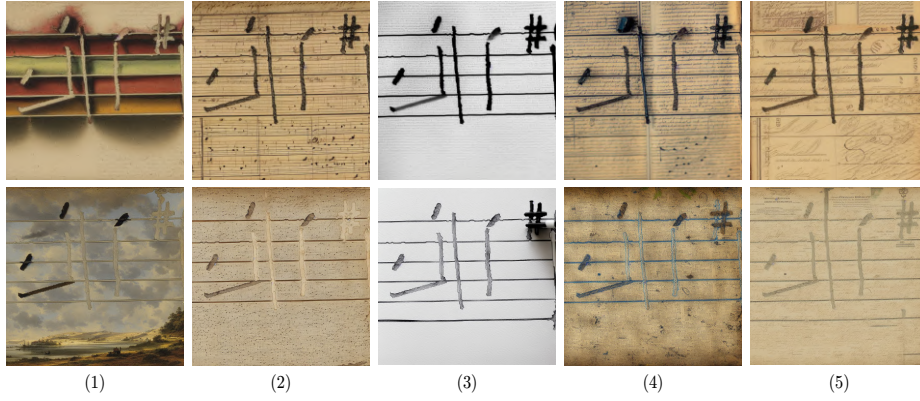


Fig. 2. Output of a ControlNet 1.1 on a crop of a Mashcima-engraved score using a 1:1 aspect ratio using diverse prompts. The background texture is decorrelated from the control signal, which we attribute to the fact that not many close-up images of music were used during training of these Models.

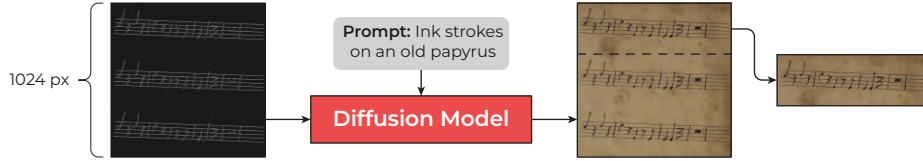


Fig. 3. Pipeline of the stitching process. Given the Canny edge mask of a 1024×1024 px image, we pass it through a diffusion model with the prompt *Ink strokes on an old papyrus*. The first occurrence of the generated image is then cropped to get the final image.

the output [28]. This can also be perceived when operating these models at odd image aspect ratios, which is fundamental for generating lines of text and music. We use the MAD operator from Quattrini *et al.* [29], which uses an improved feature combination algorithm that allows generation of more consistent wider images without needing to retrain the model on higher resolution images. Results using this technique are referred to as **MAD**.

Stitching. Using MAD to build very large images incurs on a very high computational cost. We devise a simpler technique to prevent issues such as the model not understanding compositions involving cropped music scores (as seen in Figure 2) that entails filling the default square resolution of the backbone diffusion model in such a way that it mimics the look of a full-page real score. Specifically, we repeat and concatenate together copies of the same line of music such that it reaches a square aspect ratio and we scale it to 1024px wide. We refer to this strategy as **Stitched**, and it is summarised in Figure 3.

4 Experiments

4.1 Considered Metrics

In order to evaluate the fidelity and realism of the generated score images, we propose using an array of metrics, each tailored to quantify specific qualities of their content. Note that, for all these metrics, the lower the value, the better.

Fréchet Inception Distance (FID) [30]. Perhaps the standard evaluation metric in the field of Image Generation, it measures the Fréchet distance between the features of a set of generated images and a set of reference images extracted using a standard Inception CNN. It should be noted however that this metric usually employs a network trained on natural images, which is not ideal for the analysis of images of text. To better consider the properties of images containing text, we also include a version of this score computed on binarised images (BFID). We also provide **Kernel Inception Distance (KID)** [31] and its **Binarised version (BKID)**, which try to solve some of FID’s biases towards the number of samples during evaluation.

Handwriting Distance (HWD) [32]. This metric was developed as an alternative to FID for the context of Handwritten Text Generation. It is computed as the distance between the average of features of non-overlapping patches of text images extracted using a VGG16 trained on a font classification task.

Learned Perceptual Image Patch Similarity (LPIPS) [33]. This score captures the perceptual consistency of the image by comparing all possible pairs of non-overlapping squared patches in the image under consideration against a reference image. The score is obtained as the average distance between features extracted from the patches by using a VGG16 trained on a font classification task. For readability, we report the values of this metric scaled by 10^4 .

Δ -Intra-LPIPS. This score is obtained as the difference between the Intra-LPIPS – the LPIPS score computed between all patches of a single source image – of a reference image and a generated one. This way, it captures the realism of the generated image in terms of style preservation. For readability, we report the values of this metric scaled by 10^4 .

4.2 Compared Synthetic Data Variants

The goal of this study is to evaluate the quality and usefulness of synthetic music score images when generated with the proposed strategy. To this end, we first have to consider the overall appearance of the synthetically-generated images against real, handwritten images. Thus, a set of 4,143 real, historical, handwritten, monophonic music lines are collected as reference (which we refer to as **Real Lines**). In order to minimise the domain shift caused by the musical content of these scores, their transcriptions are used to produce two “parallel” synthetic datasets with the aforementioned Mashcima engraving strategy (which we refer to as **Mashcima** or **M**) and augmentations (which we refer to as **Augmented**

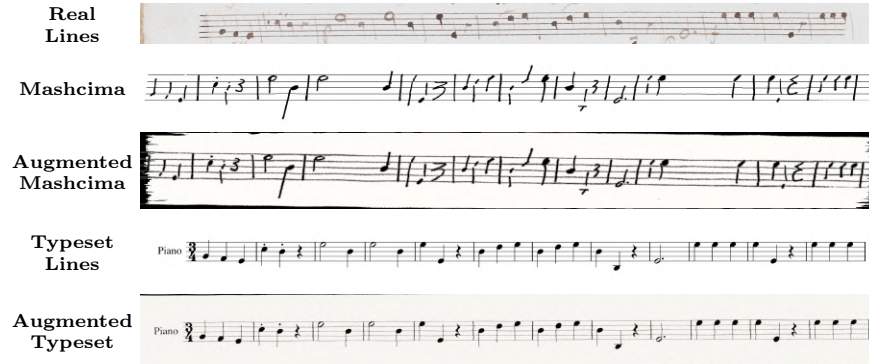


Fig. 4. Exemplar line from the considered datasets variants: the real line from the reference dataset, its Mashcima-engraved version, its augmented counterpart, the typeset version, and its augmented counterpart.

Table 1. Perceptual results evaluating on the images generated using both techniques and evaluating at a fixed height of 384px. The dash "-" means the dataset is evaluated without diffusion. For those cases where diffusion is used, results are averaged across all 5 prompts. Values for Δ -Intra-LPIPS are scaled by 10^4 .

Dataset	Method	FID	BFID	KID	BKID	HWD	LPIPS	Δ -Intra-LPIPS
Mashcima	-	200.8	174.9	0.227	0.168	2.27	3.88	-1.36
	MAD	145.1 $\pm$ 12.87	155.4 $\pm$ 15.0	0.144 $\pm$ 0.015	0.146 $\pm$ 0.017	1.99 $\pm$ 0.236	1.37 $\pm$ 0.399	-2.90 $\pm$ 0.41
	Stitched	144.0 $\pm$ 8.13	175.4 $\pm$ 7.7	0.134 $\pm$ 0.009	0.160 $\pm$ 0.012	1.82 $\pm$ 0.188	0.49 $\pm$ 0.095	-3.33 $\pm$ 0.27
Augmented Mashcima	-	132.0	139.5	0.133	0.117	2.19	1.52	-1.86
	MAD	119.6 $\pm$ 12.35	132.9 $\pm$ 3.6	0.109 $\pm$ 0.017	0.115 $\pm$ 0.005	2.01 $\pm$ 0.199	1.44 $\pm$ 0.373	-2.99 $\pm$ 0.39
	Stitched	134.5 $\pm$ 6.36	161.8 $\pm$ 5.4	0.126 $\pm$ 0.005	0.152 $\pm$ 0.009	2.10 $\pm$ 0.219	0.54 $\pm$ 0.136	-3.17 $\pm$ 0.36
Typeset	-	192.3	217.8	0.192	0.196	1.74	1.29	2.21
	MAD	166.2 $\pm$ 21.79	221.5 $\pm$ 13.0	0.164 $\pm$ 0.029	0.214 $\pm$ 0.018	2.16 $\pm$ 0.421	1.60 $\pm$ 0.559	-0.52 $\pm$ 1.42
	Stitched	155.4 $\pm$ 6.54	186.0 $\pm$ 7.1	0.136 $\pm$ 0.009	0.157 $\pm$ 0.007	1.21 $\pm$ 0.136	0.46 $\pm$ 0.046	-0.47 $\pm$ 1.83
Augmented Typeset	-	125.8	117.9	0.114	0.086	1.60	0.65	0.13
	MAD	141.3 $\pm$ 12.65	188.4 $\pm$ 13.1	0.136 $\pm$ 0.015	0.179 $\pm$ 0.013	2.04 $\pm$ 0.284	1.43 $\pm$ 0.292	-1.41 $\pm$ 1.08
	Stitched	152.6 $\pm$ 7.86	184.4 $\pm$ 9.8	0.138 $\pm$ 0.006	0.162 $\pm$ 0.012	1.64 $\pm$ 0.321	0.48 $\pm$ 0.073	-0.87 $\pm$ 1.58

Mashcima or AM). We also make sure to match the paper texture of the generated images to the average background color of the real ones in order to be able to evaluate the results both before and after feeding them to the diffusion model more fairly, as we expect the background to play a role in any metric we employ. For the sake of comparison, we repeat this process using Verovio [34] as the engraver for a typeset style, thus obtaining the **Typeset** or **T** and **Augmented Typeset** or **AT** variants. In Fig. 4 we report an exemplar line within the different variants.

4.3 Generation Results

In this section, we discuss results with the Stitched approach and with the MAD-based approach, with quantitative results in Table 1 averaged by prompt.

As we can see, diffusion-based generation generally improves FID and KID scores, especially for the non-augmented results. On the perceptual metric LPIPS, we see consistent improvement for both Stitched and MAD approaches compared to the baseline. On the other hand, we see mixed results on the content-oriented metrics (BFID, BKID, and HWD), with scores not consistently improved by the generation process. Overall, these results show that diffusion-generated images are more realistic and aesthetically similar to the original samples than the engraved images, but not by a very wide margin.

For HWD, we qualitatively noticed that it tends to favour images that contain unrelated text in the background, generated by the diffusion model. We observed this most prominently on Prompt 0 (see Appendix [D](#)), which tends to generate background text. We can also notice that the non-diffusion Augmented Typeset baseline has solid performance on most metrics. To discard this being caused by the background color matching operation, we also experiment with altering the hue in these Augmented Typeset baseline images by a random factor, resulting in minimal increments on FID and KID of +2.1 and +0.001 respectively.

In Fig. [5](#) we provide qualitative samples for our experiments, comprising the MAD and Stitching-based generation on the AM and AT samples. In general, we can see that all the proposed prompts variations generate realistic background textures and ink effects. Another aspect that emerges from the quantitative evaluation and that is also evident in these samples is that the content is not much affected by the diffusion process, as ControlNet poses strong constraints on the symbols variability. Therefore, the resulting images have realistic backgrounds and ink effects, yet with the same handwriting style as the input source images. Looking at the MAD samples in the top portion of Fig. [5](#), we see that this generation approach is more sensitive to the textual prompt. Especially for prompts 0 and 1, the model tends to fill the background with gibberish text and scribbles, trying to reproduce the appearance of the full-page images that are part of the large-scale training dataset of Stable Diffusion. This effect can be alleviated by choosing more neutral prompts (*e.g.*, prompts 3 and 4).

We have also observed that it is possible to further influence the quality of the output through clever use of data augmentation. Some operations on the input images seem to induce convincing handwritten artifacts during the diffusion process: Gaussian noise, which the model in many cases converts to some form of Kanungo noise effect in the output image, and paper cropping effects at the sides of the score, strengthening the idea that the image contains a piece of paper.

4.4 OMR Results

We experimented using an end-to-end OMR system on line-level samples to check whether using generated samples alone substitutes using the real ones. We trained an HTR-VT [\[35\]](#) model on some splits of the generated samples — more specifically, those that seemed to perform best in terms of perceptual metrics using both MAD and Stitched strategies, which were based on the Augmented Mashcima dataset — and compared their results with the default Typeset and

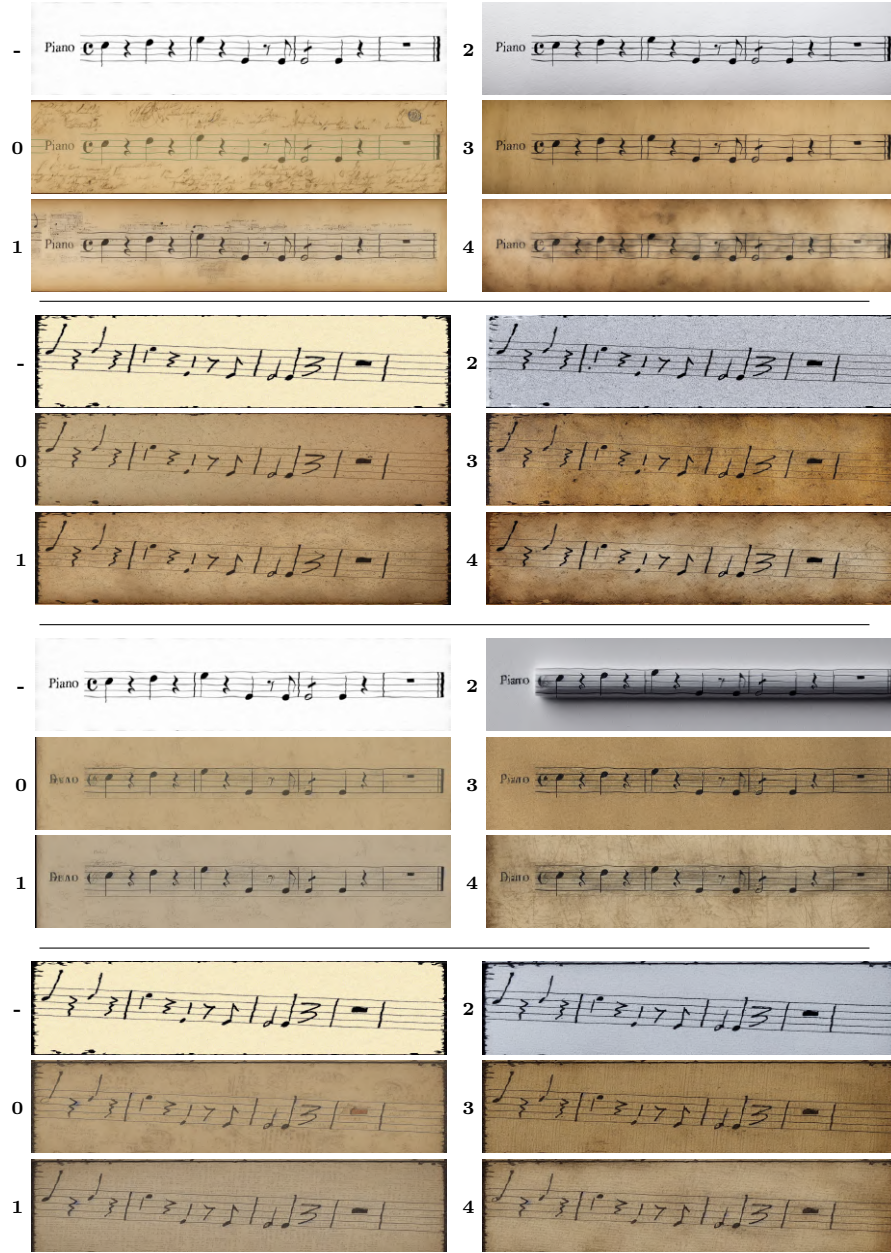


Fig. 5. Collection of qualitative examples produced from the same source transcription using all techniques and prompts. From top to bottom, samples produced using MAD on Augmented Typeset (AT) samples, MAD on Augmented Mashcima (AM) samples, Stitched on AT and Stitched on AM. Prompt IDs shown on the left.

Table 2. OMR SER results using an HTR-VT model on real, regular and augmented typeset samples and diffusion-based augmented Mashcima samples tested on real lines.

	No Diffusion			Diffusion	
	<i>Real</i>	<i>T</i>	<i>AT</i>	<i>MAD-AM</i>	<i>Stitched-AM</i>
SER	0.23	0.65	0.63	0.63	0.64
Δ-SER	-	0.42	0.41	0.40	0.41

Augmented Typeset samples. Note that the training transcriptions are the exact same for all datasets in order to isolate the effect of number of samples and transcription domain shift. All generated samples are created from the transcriptions of the train split of the real dataset. Results are shown in Table 2 using Symbol Error Rate, which is the number of Levenshtein edits to obtain the ground truth sequence from the result divided by the length of the ground truth. From this experiment we conclude that training on the generated samples alone is not better than using normal data augmentation on the regular typeset samples if the goal is improving results on real lines.

5 Conclusions

In this paper, we have explored the generation of handwritten music scores by using off-the-shelf ControlNet models. To the best of our knowledge, this is the first work that addresses the task of Optical Music Generation with this level of granular control over the semantic content of the music scores. Both techniques presented here — MAD and Stitched — seem to partially bridge the gap between real and computer-generated scores, but we have seen experimentally that they are not quite good enough on their own to improve results on OMR tasks directly as they are not perfect replacements of real training samples. They can, however, be useful as pre-training or curricular samples in a larger training process, preventing overfitting to an overly-clean score style.

There are some open issues that need to be further explored for the application of the proposed approach in training Handwritten Music Recognition systems. Mainly, the results are very prompt-dependent, and we leave for future works the exploration of a consistently reliable way to use a style reference to generate images. Secondly, some results still suffer from artifacts, especially with prompts that refer to more abstract concepts, such as documents or scores. However, this is mostly due to the underlying Stable Diffusion 1.5 model, and this limitation will be alleviated with the development of increasingly large-scale image generation models. It might be interesting to approach this task as an image editing one as well. Finally, this work highlights the need for a metric tailored for music score generation, as no current metric satisfactorily measures the full set of graphical properties that can be found within a handwritten music score.

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